SE4042 Modelling and Simulation: Lab 6, Finite Element Methods

Name: M.C.R.Mendis Student ID: IT23173118 Date: 28 October 2026

Part A: The whole pipeline on five nodes

Predict first: without the heat source, what would T5 be? With the source, higher or lower, and by how much?

Without the source T5 = 60 °C: node 5 is linked with equal weight to the four corners, so it takes their average, (100 + 20 + 20 + 100) / 4 = 60 °C. With the source T5 is higher, because the generated heat has to flow out through the edges. The rise is f5 / K55 = 333.33 / 40 ≈ 8.3 °C, giving about 68.3 °C.

A1: By hand

Quantity	Your value
Ke (numbers)	(k t / 2) = 5, so Ke = [5 0 -5; 0 5 -5; -5 -5 10]
fe (numbers)	Q A t / 3 = 1000 × 0.25 × 1 / 3 = 83.33, so fe = [83.33; 83.33; 83.33]
K55 and K5j (j = 1...4)	K55 = 4 × 10 = 40; K51 = K52 = K53 = K54 = 2 × (-5) = -10
f5	4 × 83.33 = 333.33
Equation for T5	40 T5 - 10 (100 + 20 + 20 + 100) = 333.33, so 40 T5 = 2400 + 333.33
T5	68.33 °C

A3: Questions

1. Did your T5 agree with the program? If not, which step differed first?

Yes. The program gives T5 = 68.33 °C (Learn, Step-by-step, step 9), the same as the hand calculation. Every intermediate step also matched: element 1 has Ke = [5 0 -5; 0 5 -5; -5 -5 10] and fe = 83.33 per node (step 2), K55 = 40 with off-diagonals -10 (step 6), and f5 = 333.33 (step 7).

2. In K, the entry linking nodes 1 and 2 is zero although they share element 1. Why?

For a linear triangle the off-diagonal entry for edge ij is -(k t / 2) cot θ, where θ is the angle opposite that edge. Edge 1-2 is opposite node 5, where the triangle has its right angle, and cot 90° = 0. So K12 = 0 even though both nodes are in element 1.

3. Step 2 says each Ke has one zero eigenvalue. Which temperature field does it correspond to, and why must it cost no energy?

The eigenvector is the uniform field T1 = T2 = T5 = constant. A constant temperature has zero gradient, so there is no heat flux and the energy ½ T^T Ke T is zero. Adding the same constant to every temperature cannot change the conduction, which is why Ke on its own is singular until a temperature is fixed.

4. Heat flow at fixed-temperature boundary: write the two values and show that they balance the heat generated.

Hot side (100 °C): +300 W (into the plate). Cold side (20 °C): -1300 W (out of the plate). Generated heat = Q × area × t = 1000 × 1 × 1 = 1000 W. Balance: 300 + 1000 = 1300 W leaves through the cold edge, and the program reports a residual of 0 (balanced).

Part B: Springs in disguise, a three-bar truss

Bar lengths: L12 = 10 m, L23 = 10 m, L13 = 10√2 = 14.142 m. E = 200 GPa.

Reduced system from Learn, Step-by-step, step 7 (unknowns ux2, ux3, uy3):

K_ff = [2.000e7 0 0; 0 2.000e7 2.000e7; 0 2.000e7 3.000e7] N/m, f_f = [0; 20000; 10000] N

Solution by hand: row 1 gives 2e7 ux2 = 0, so ux2 = 0. Subtracting row 2 from row 3 gives 1e7 uy3 = 10000 - 20000, so uy3 = -1.000 mm. Then ux3 = (20000 - 2e7 × (-0.001)) / 2e7 = 2.000 mm. Bar 1-3 force: c = (0.7071, 0.7071), so N13 = 3.9994e7 × (0.7071 × 0.002 + 0.7071 × (-0.001)) = 28.28 kN (tension).

	Bar 1-2	**Bar 2-3**	**Bar 1-3**
EA/L (N/m)	2.000×10^7	1.000×10^7	3.999×10^7
N from your hand calculation (kN)	0	-10.00 (compression)	+28.28 (tension)
N from Results, Member forces (kN)	0	-10	28.28

	ux2	**ux3**	**uy3**
Your solution of K_ff u_f = f_f (mm)	0	2.000	-1.000

Check: the program reports max |U| = 2.236 mm = √(2.000² + 1.000²) at node 3, and the support reactions Rx1 = -20 kN, Ry1 = -20 kN, Ry2 = +10 kN balance the load.

4. Bar 1-2 carries zero force. Explain why from the free-body diagram of node 2.

At node 2 only three things act: bar 1-2 (horizontal), bar 2-3 (vertical) and the roller (vertical reaction only). There is no external horizontal load at node 2, so horizontal equilibrium contains only the force of bar 1-2, which must therefore be zero. This is also why ux2 = 0.

5. Delete the roller at node 2 and read the pre-flight check. Which motion is it warning about, and what does that mean for K?

The pre-flight check says: "The model can still move freely: rotation about z (the part can pivot about its support)." With only the pin at node 1, the whole truss can rotate about node 1 as a rigid body. That motion produces no bar strain, so K_ff has a zero eigenvalue: it is singular and cannot be factorised or inverted. Undo (Ctrl+Z) brought the roller back.

Part C: Your own model, the L-shape

Predict first: when you halve the element size, what should happen to (a) max T and (b) the largest heat flux |q|?

(a) Max T should converge, because u ~ r^(2/3) is continuous and bounded, so the maximum settles to a limit. (b) The largest |q| should keep growing, because |∇u| ~ r^(-1/3) is infinite at the corner. Halving h samples the field half as far from the corner, so the peak should grow by about 2^(1/3) ≈ 1.26 each time and never converge.

Checkpoint: why exactly 0.03 W?

In steady state all generated heat must leave through the fixed-temperature edges: area × Q × thickness = 3 m² × 1 W/m³ × t. With t = 0.01 m this gives 0.03 W. In the version of FEM Studio I used, the default thickness in 1 · Problem was 1 m, so the table showed -3 W (149 nodes, T range 0 to 0.146 as expected). The temperatures do not depend on t.

C2: Convergence study

Element	h (m)	Nodes	max T	max |q| (marker)	ZZ error (%)
T3	0.2	149	0.1462	0.595	28.9
T3	0.1	583	0.1481	0.720	16.3
T3	0.05	2204	0.1490	0.984	9.1
T3	0.025	8744	0.1492	1.239	5.2
T6	0.2	548	0.1490	0.965	9.8
T6	0.1	2238	0.1491	1.115	7.4
T6 (optional)	0.05	8630	0.1493	1.487	3.9
T6 (optional)	0.025	34617	0.1494	1.859	2.5

1. Does max T converge? To what value? Does max |q| converge?

Yes, max T converges: 0.1462, 0.1481, 0.1490, 0.1492 (T3) and 0.1490 to 0.1494 (T6), settling at about 0.1494. Max |q| does not converge: it grows on every refinement, from 0.595 to 1.239 (T3) and from 0.965 to 1.859 (T6), and the peak always sits at the re-entrant corner (1, 1).

2. Compute the ratio of max |q| between successive rows (T3). Compare with 2^(1/3) ≈ 1.26. What would you tell an engineer who reports the peak flux at the corner?

T3 ratios: 0.720 / 0.595 = 1.21, 0.984 / 0.720 = 1.37, 1.239 / 0.984 = 1.26 (mean 1.28), close to 2^(1/3) = 1.26, so the peak grows like h^(-1/3) as predicted. I would tell the engineer that the corner value is not a result: the exact flux there is infinite, and the number only reflects the element size chosen. They should report the flux a short distance from the corner (where it does converge), or model the real geometry, for example a fillet radius.

3. At equal h, T6 has about four times as many nodes as T3. Is T6 worth it here? Use the ZZ column.

Only modestly. Compared at a similar number of nodes, T6 does have a smaller ZZ error: 9.8 % against 16.3 % (about 550 nodes), 7.4 % against 9.1 % (about 2200), and 3.9 % against 5.2 % (about 8700). But the gain is small and shrinks, because the corner singularity limits both elements to the same rate. On a smooth problem T6 would be far better than T3 for the same nodes; here refining at the corner is worth much more than raising the element order.

C3: Spend the nodes where the error is

Mesh	Nodes	ZZ error (%)
T6, h = 0.2, refined to 0.01 at (1, 1)	2653	2.3
T6, h = 0.1 uniform (from C2)	2238	7.4
T6, h = 0.025 uniform (from C2)	34617	2.5

4. Compare the graded mesh with uniform T6 h = 0.1. How many times fewer nodes does the graded mesh need? Which idea from the lecture does this imitate?

With a similar number of nodes (2653 against 2238) the graded mesh has about one third of the error (2.3 % against 7.4 %). It reaches the accuracy of uniform h = 0.025 (2.5 %, 34617 nodes) with about 13 times fewer nodes. This imitates adaptive mesh refinement from the lecture: put small elements where the error indicator is large (the singular corner) and large elements where the solution is smooth.

C4: The graded mesh was exported as L_T6_graded.vtk (with L_T3_0.1.vtk and L_T6_0.2.vtk) and the model was saved as L-shape.fem.json.

Part D: Can you trust the answer?

D1: Stress concentration at a hole (Kirsch, Kt ≈ 3)

Predict first: which element (T3 or T6) will get close with fewer nodes, and will nodal averaging read high or low?

T6 should get close to 300 MPa with fewer nodes, because its stress varies linearly inside each element while T3 stress is constant per element. Nodal averaging should read low: averaging smears the peak at the hole edge with the lower values of the neighbouring elements.

Size there (mm)	T3: nodes	T3: max vM avg / SPR (MPa)	T6: nodes	T6: max vM avg / SPR (MPa)
4	444	198.8 / 198.6	1664	281.9 / 264.8
2	661	250.1 / 274.8	2454	299.3 / 304.9
1.2	944	292.1 / 306.0	3484	303.2 / 305.5
0.6 (optional)	1627	294.3 / 302.7	5971	303.3 / 304.0

1. Which element converges faster? Why does the averaged T3 value stay low even when the mesh is fine?

T6 converges faster: it is already within 1 % of its final value (about 303 MPa, slightly above 300 because the plate has a finite width) at 2 mm, and is converged at 1.2 mm. Averaged T3 is still 294 MPa at 0.6 mm. A T3 element has a linear displacement field, so its strain and stress are constant inside the element. The constant value belongs to the element centroid, which lies away from the hole edge, and nodal averaging then mixes it with neighbouring elements, so the peak is always underestimated. SPR recovery (302.7 MPa) corrects most of this.

2. Watch the Max displacement |U| tile while you refine. What does that tell you about displacements versus stresses?

Max |U| hardly changes: 0.1020 to 0.1021 mm for every mesh and both elements, while the peak stress changed by 50 %. Displacements are the primary unknowns and converge one order faster than stresses, which are derivatives of the displacement and converge more slowly (Chapter 6, §6.7). A displacement that has converged does not mean the stress has.

D2: The bracket corner, a number that is not a result

Size there (mm)	max vM avg (MPa)	max vM SPR (MPa)	max |U| (mm)	Probe ≈ (50, 30) (MPa)
4	269.0	187.8	0.6744	114.7
1	467.6	304.4	0.6780	114.7

0.25	997.5	613.9	0.6789	114.7
2 (optional)	333.2	216.3	0.6765	114.7
0.5 (optional)	639.4	434.6	0.6786	114.7

3. Which columns converge and which do not? Read the yellow warning and connect it to the L-shape in Part C.

Max |U| (0.674 to 0.679 mm) and the probe stress at (50, 30) mm (114.7 MPa every time) converge. Both peak von Mises columns, averaged and SPR, keep growing without limit. The yellow warning says the maximum von Mises stress is at (30, 30) mm, at a re-entrant corner with an interior angle of about 270°, where the exact stress is infinite (a singularity), so the peak keeps growing as the mesh is refined and must not be read as the strength-relevant stress. This is the same corner as the L-shape in Part C: there the heat flux |q| diverged at (1, 1); here the stress does.

4. Each step from 4 to 1 to 0.25 mm divides the corner element size by 4. By what factor does the peak grow per step? What does σ ~ r^(-0.46) predict? What should a designer do instead?

Measured growth of the averaged peak: 467.6 / 269.0 = 1.74 and 997.5 / 467.6 = 2.13 (geometric mean 1.93). Theory: σ ~ r^(-0.46) predicts 4^0.46 = 1.89 per step, in good agreement. A designer should not read the peak. Instead they should model the real fillet radius and run a convergence study on it, read the stress a fixed distance from the corner (as the probe does), or use a method meant for singular points (notch or fracture mechanics).

D3: Locking, the element that cannot bend

Element (formulation)	**Divisions x × y**	**Tip deflection (mm)**	**% of 2.00 mm**
Q4 (Standard 2 × 2 Gauss)	10 × 1	1.362	68.1 %
Q4 (Standard)	10 × 4	1.444	72.2 %
Q4 (Standard) (optional)	40 × 4	1.956	97.8 %
Q4 (Incompatible modes QM6)	10 × 1	2.014	100.7 %
Q8 (optional)	10 × 1	2.002	100.1 %

5. Refining through the thickness (10 × 4) barely helps the standard Q4. Why can a bilinear element not bend without shearing, and what does QM6 add to fix it?

A bilinear Q4 element keeps its edges straight, so it cannot take the curved shape of pure bending. To follow a bending moment it has to produce a spurious shear strain (parasitic shear) that a real beam does not have. That shear absorbs strain energy and makes the element far too stiff (shear locking). Adding layers through the thickness does not help, because every element is still long and thin and still locks; only shortening the elements (40 × 4) reduces it. QM6 adds internal incompatible bending modes, $(1 - \xi^2)$ and $(1 - \eta^2)$, to each displacement. These let the element edges curve so it can bend without spurious shear, and they are condensed out at element level. One layer then gives 100.7 % of the beam value.

Part E: Break the element on purpose

	Your observation
Validate, original code	All checks pass: Compiles; symmetric Ke and rigid-body modes (one zero eigenvalue, a constant field) for T3, T6, Q4 and Q8; patch test T3 and patch test Q8 (linear field, max error about 1e-14).
Validate without det J: checks that pass	Compiles; symmetric Ke for T3, T6, Q4, Q8; rigid-body modes (1 zero eigenvalue) for T3, T6, Q4, Q8.
... checks that fail	Patch test T3 (linear field), max error 1.6; patch test Q8 (linear field), max error 1.4.
T5 with the broken element	64.17 °C (instead of 68.33 °C); the heat flows became 1100 W (hot) and -2100 W (cold).

6. Why do the symmetry and zero-eigenvalue checks still pass when every Ke has been multiplied by a wrong positive factor?

Multiplying a matrix by a positive constant c does not change its structure: if $Ke = Ke^T$ then $c\,Ke = (c\,Ke)^T$, and if $Ke\,T = 0$ for a constant field T then $c\,Ke\,T = 0$ as well. The eigenvalues are only scaled, so the same single zero eigenvalue remains. These checks test the form of Ke, not its size.

7. On the 4-triangle model the broken element gives exactly the same T5 as doubling k would. Why? Why does the patch test on an irregular mesh catch the bug when this test problem cannot?

All four triangles are identical, with det J = 2A = 0.5. Removing det J multiplies every Ke by the same factor 1/0.5 = 2, so the global K doubles while f is unchanged. That is exactly what k = 20 would do: T5 = (2400 × 2 + 333.33) / 80 = 64.17 °C. The answer is wrong but physically plausible, so this problem cannot reveal the bug. In the patch test the elements have different sizes and shapes, so each one is scaled by a different 1/det J. This behaves like a conductivity that jumps from element to element, so the mesh can no longer reproduce an exact linear field, and the patch test fails.

Part F: Python, check the numbers yourself

P1: Three errors, three rates (1-D)

n	**L^2 error**	**energy error**	**nodal error**
5	2.520e-02	4.003e-01	1.051e-04
10	6.353e-03	2.011e-01	6.800e-06
20	1.592e-03	1.007e-01	4.233e-07
50	2.548e-04	4.029e-02	1.083e-08
100	6.371e-05	2.015e-02	6.765e-10
slope vs n	-2.00	-1.00	-3.99

Predict first: P1c uses a one-point rule for the load integral. Which of the three slopes will change, and to what?

Only the nodal slope should change. The one-point rule adds a load-integration error of order h^2, which destroys the very high nodal accuracy, so the nodal slope should drop from about -4 to -2. The

L^2 (-2) and energy (-1) slopes are set by how well piecewise-linear functions approximate sin πx, so they should stay the same.

P1c (nq = 1)	L^2 slope	energy slope	nodal slope
slope vs n	-2.00	-1.00	-2.00

1. Which slope matches the theory for linear elements in each norm? Which one is the nodal trap, and what did P1c do to it?

For linear elements the theory gives $O(h^2)$ in L^2 and $O(h)$ in the energy norm, and the measured slopes -2.00 and -1.00 match exactly. The nodal error is the trap: in 1-D the finite element solution is (almost) exact at the nodes, so the nodal error falls like h^4 (slope -3.99) and makes the solution look far more accurate than it really is between the nodes. P1c removed the illusion: with a one-point load rule the nodes are no longer exact and the nodal slope fell to -2.00, while the L^2 and energy slopes did not change.

P2: The L-shape in code

m	nodes	max u	energy error
4	65	0.1372	0.1580
8	225	0.1459	0.0862
16	833	0.1481	0.0476
32	3201	0.1490	0.0269
64	12545	0.1493	0.0156

Slope vs N: -0.412 Slope vs h: 0.824

Smooth-solution theory for P1 would give a slope of 1 against h. Why is yours smaller, and what value is it heading for?

The re-entrant corner makes the exact solution singular: u ~ r^(2/3), so its gradient is not square-integrable to high order and no uniform mesh can approximate it at the smooth rate. The energy error is limited to O(h^(2/3)) regardless of the element. The local slopes between successive meshes keep falling (0.87, 0.86, 0.82, 0.79), so the fitted 0.82 is still heading down towards 2/3 ≈ 0.67 (that is, -1/3 against N).

P3: Exact errors for FEM Studio's meshes

FEM Studio mesh	nodes	∫uh	energy error
T3, h = 0.1	583	0.211421	0.0515
T6, h = 0.2	548	0.213541	0.0231
T6 graded (C3)	2653	0.214038	0.0062

2. Compare the three energy errors, taking the node counts into account. Which mesh gives the most accuracy for its size? Does the ZZ estimate rank them the same way?

T3 h = 0.1 and T6 h = 0.2 have almost the same number of nodes (583 and 548), and T6 has less than half the error (0.0231 against 0.0515). The graded T6 mesh is by far the most efficient: with 2653 nodes its error is 0.0062, about 4 times smaller than T6 h = 0.2 and 8 times smaller than T3 h = 0.1. For comparison, the uniform P2 mesh with 3201 nodes still has 0.0269. The ZZ estimate ranks them

the same way (16.3 %, 9.8 %, 2.3 %), although it overestimates the true relative error, which is 11.1 %, 5.0 % and 1.3 % (energy error divided by √0.21407581). On the log-log plot (P3b) the graded point lies far below the uniform line, and it is the only one that beats the corner-limited slope of -1/3 against N.

Exit questions

1. Name one quantity in today's lab that converged and one that did not. What made the difference?

Converged: max T on the L-shape (about 0.1494), and the displacement and probe stress on the bracket (0.679 mm, 114.7 MPa). Did not converge: the peak heat flux at the L-shape corner and the peak von Mises stress at the bracket corner. The difference is whether the exact quantity is finite. At a re-entrant corner the gradient (flux or stress) of the exact solution is infinite, so its maximum grows with every refinement. Quantities that are finite (the solution itself, integrals, or values away from the corner) converge.

2. A classmate says: "My T6 mesh gives an L^2 error slope of 3 on a smooth problem, so it will give slope 3 on any problem." Use Part C or P2 to correct them.

The rate O(h^(p+1)) only holds when the exact solution is smooth enough. On the L-shape the corner singularity u ~ r^(2/3) limits every element order to the same reduced rate. In P2 the energy error slope against h was only 0.82 (heading to 2/3) instead of 1. In Part C, uniform T6 cut the ZZ error by only about 1.3 to 1.9 times per halving of h, far from the factor of 4 that a smooth problem would give. Higher order cannot beat a singularity; only local refinement at the corner restores the accuracy.

3. List three checks you would make before trusting an FE stress result in a design report.

(1) An equilibrium check: the reactions balance the applied loads, and the model is properly supported (no rigid-body motion). (2) A mesh convergence study of the reported stress, refining where the gradient is high and checking that the averaged and SPR values agree; a peak that keeps growing is a singularity, not a result. (3) A check against an independent estimate (a hand calculation such as Kt ≈ 3 or beam theory) and of the element choice (for example no shear locking from Q4 in bending), with stresses read away from singular corners, point loads and support edges.